

CMSC6950 — Fall 2025

Homework #4

Due Date: Work must be submitted to GradeScope (for Problems 1-2) or GitHub (for Problem 4) by 5pm (St. John's time) on Thursday, November 6.

1. (20 points) Download the `gapminder` data from the course page. For each of the files, load the data and use the `quantile` method from `pandas` to compute the 25% and 75% quantile of the data for the associated continent. For each year of data, define a country to be “wealthy” if their GDP per capita exceeds the 75% quantile, to be “poor” if it is less than the 25% quantile, and to be “middle” if it is between the two. For each continent and each year of data, output the mean GDP per capita for each of these three groups. *Hint:* For the “middle” countries, it may be useful to know that you can mask based on a logical and statement, but must use `&` instead of `and` to do this.
2. (20 points) For each of the following tasks, state whether a `for` or `while` loop would be a better solution for implementing the task and explain why.
 - (a) Determining how many numbers in a given list are odd.
 - (b) Finding the fifth odd number in a given list (if there are at least five such numbers in the list)
 - (c) Finding the first time the product of two consecutive numbers in a list is greater than 20 and at least one number is positive (if this occurs in the list)

Write code for one of the tasks (including outputting the number(s) found) and demonstrate it on a small data set of your choice.

3. (20 points) Go to <https://classroom.github.com/a/wHDOZpcz> and accept the “HW4 Lists” assignment. Clone the repository, read the instructions in the `README.md` file, complete the assignment, and push your code to the repo to be graded. Nothing needs to be handed in to GradeScope for this problem.